

BIOLOGY 30 FINAL REVIEW

Unit 1: What is Life?

Exploring the scientific, cultural, and historical definitions of the biological world.

Section 1.1

The Scientific Criteria

Defining life through biological hierarchy and metabolism.

| THE 7 CHARACTERISTICS

1. Order: Organized cellular hierarchy (Atoms → Cells → Tissues).

2. Reproduction: Passing genetic material to offspring.

3. Growth & Development: DNA-guided maturation process.

4. Energy Processing: Transforming food/sunlight into energy (Metabolism).

5. Response to Stimuli: Reacting to environment (e.g., Venus Flytrap).

6. Homeostasis: Keeping internal states stable despite external changes.

7. Evolutionary Adaptation: Population-wide changes over generations.

Section 1.2

Diverse Perspectives

Bridging Western Reductionism and Indigenous Holism.

WESTERN REDUCTIONISM

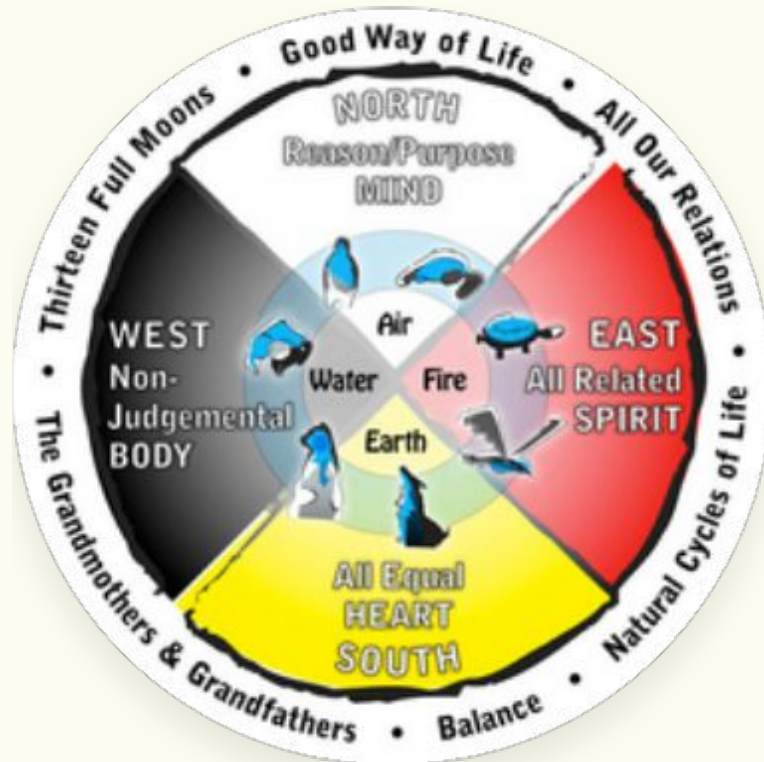
The Western scientific approach typically seeks to understand complex biological systems by **breaking them down into smallest parts.**

By studying organelles, proteins, and atoms in isolation, scientists aim to piece together the mechanics of the "whole machine."

This perspective is often mechanistic, driven by the laws of chemistry and physics.



INDIGENOUS HOLISM



Wâhkôhtowin (Kinship)

Indigenous worldviews emphasize a **web of relationships** connecting all things, animate and inanimate. Life is defined by connection, not isolation.

The concept of *Wâhkôhtowin* teaches that we are in constant relationship with the land, water, and ancestors. Severing these ties diminishes life itself.

Section 1.3

The History of Biogenesis

The paradigm shift from spontaneous generation to living reproduction.

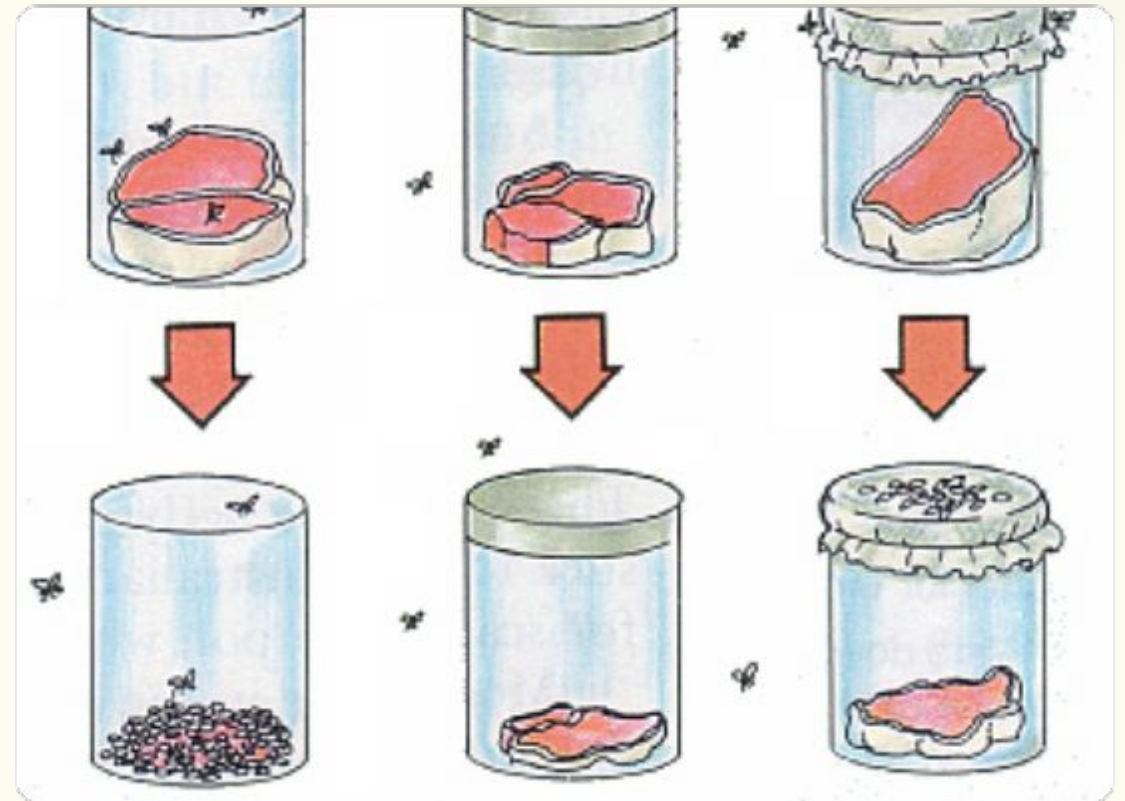
REDI'S 1668 EXPERIMENT

Challenging Spontaneous Generation

Francesco Redi used three jars of meat to disprove that maggots arise from decaying flesh:

- **Open Jar:** Maggots appeared (control).
- **Sealed Jar:** No maggots appeared.
- **Gauze Covered:** Maggots appeared on top of gauze, proving they come from fly eggs.

This proved that macroscopic life comes from pre-existing parents.



PASTEUR'S LANDMARK PROOF

1859

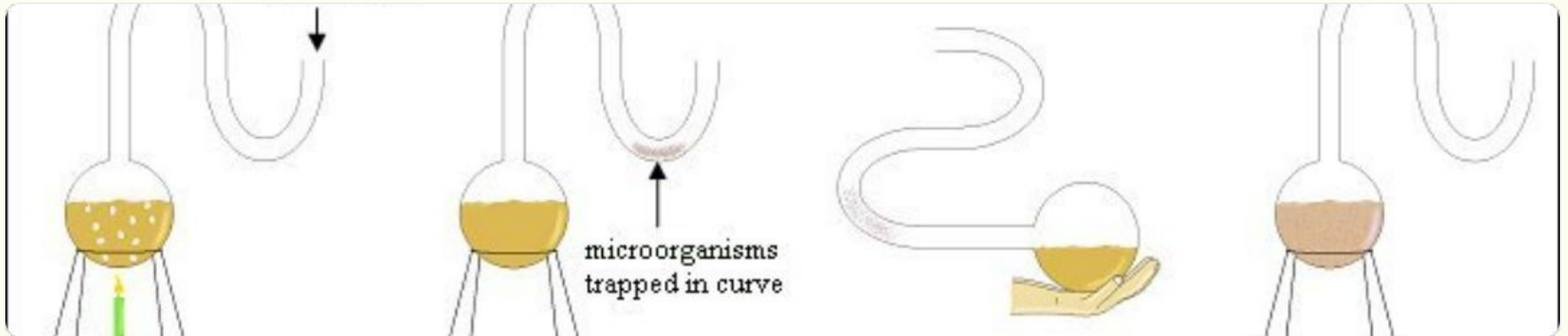
The Year of Biogenesis

The Swan-Neck Flask

Louis Pasteur proved that microbes are in the air, not created by it. His curved flask necks let air in but trapped dust/microbes.

Key Takeaway: Broth remained sterile for months until tilted.

This final proof ended the debate: *Omne vivum ex vivo* (All life comes from life).



SUB-CELLULAR GRAY AREAS

Entity	Structure	Genetic Material	Status
Viruses	Capsid (Protein)	DNA or RNA	Obligate Parasite
Viroids	Naked Circular Loop	RNA Only	Infects Plants
Prions	Misfolded Protein	None	Infectious Protein

These entities challenge our criteria by lacking independent metabolism and reproduction.



A photograph of a deer with large antlers standing in a field of dry grass. The image is darkened to serve as a background for text.

SASKATCHEWAN CONNECTION

Chronic Wasting Disease

CWD is a fatal prion disease in local deer and elk. Unlike viruses, prions persist in prairie soil for years. Saskatchewan is a global leader in tracking and researching this "non-living" infectious agent.

CELL THEORY & TECHNOLOGY

